CC

CIVICCONNECT AI

SMART CITY ISSUE REPORTING AND ROUTING

PROJECT OVERVIEW

CivicConnect AI

Project Overview

A web application for reporting civic infrastructure problems, routing them to the right department, and showing progress with a complaint ID. This overview explains the need, the workflow, the technology, and the privacy boundaries of the class demo.

REPORT

Capture a local issue

ROUTE

Prioritize and assign work

TRACK

Follow progress to resolution

CLASS PROJECT DEMO | 30 OCTOBER 2026

01 PURPOSE AND USERS

Why the project exists

Civic issues such as road damage, overflowing waste, broken streetlights, water leaks, and drainage problems are easier to address when reports reach the responsible team and residents can see what happens next. CivicConnect AI brings those steps into one web portal.

The project demonstrates a complete civic report lifecycle: a person submits an issue, the system classifies and prioritizes it, the report is routed to a department, and a public complaint timeline shows status changes. It is a class demonstration, not an official city service.

Who uses the application

ROLE	WHAT THE ROLE CAN DO	DEMO ACCESS
Guest	Browse public reports, submit anonymously, and track by complaint ID.	No account required.
Citizen	View reports linked to the demo citizen account, follow progress, and support a shared issue.	Use the protected demo persona or register with a username and password.
Municipal Admin	Review city-wide reports, analytics, department assignments, and status history.	Protected demo persona.
Department Officer	Work a department queue, add progress notes, and submit resolution proof.	Protected demo persona for a department.

Why the role separation matters

Residents need a simple way to report and follow an issue. Municipal administrators need an overview across departments. Officers need a focused work queue. The role-based portals show how the same report moves through these responsibilities.

02 REPORT LIFECYCLE

From a report to a visible outcome

OPEN – REPORT – TRACK – SWITCH – WRAP UP

Suggested walkthrough: 8 to 10 minutes

1. Submit an issue. A guest or citizen enters a short title, a description, a category, and a location. A photo can be attached. Reports do not ask for a reporter name, phone number, or email address.

2. Analyze and prioritize. The API compares issue text with civic category keywords and checks basic image features when an image is present. It combines severity, safety risk, location importance, and nearby community reports into a priority score.

3. Route to a department. Category rules recommend a team such as Roads and Transportation, Sanitation, Electricity, Water Supply, Municipal Engineering, or Emergency / Public Safety.

4. Track progress. Each report receives a complaint ID. The public tracker shows a five-step timeline and an audit history without exposing the reporter account record.

5. Complete field work. An officer can start work, add a note, and submit proof when resolving an issue. Administrators can review status and assignment changes.

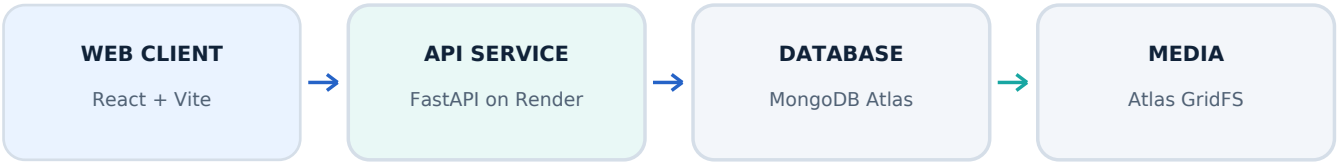
How the priority score works

FACTOR	MAX POINTS	EXAMPLE SIGNALS
Issue severity	40	Category-specific baseline and limited visual-feature adjustment.
Safety risk	25	Hazard terms such as live wire, accident, emergency, or collapse.
Location importance	15	Terms such as school, hospital, market, junction, or highway.
Community reports	20	A higher count of similar nearby reports raises the crowd-impact factor.

Priority bands are High at 70 or above, or when the safety score reaches 20; Medium from 42 through 69; and Low below 42. The score is a rule-based demonstration aid. A person must verify a real issue and decide how to respond.

03 TECHNOLOGY AND ARCHITECTURE

How the web application is assembled



HTTPS requests and JSON responses; report photos use persistent object storage in the database cluster.

The browser presents the React interface. Vite builds the frontend, which is hosted on Vercel. The interface sends API requests through the same-origin Vercel routes to the FastAPI service on Render. MongoDB Atlas stores users, complaints, and status history; production report images use Atlas GridFS so they survive an API service restart.

Main technology choices

PART	TECHNOLOGY	PURPOSE
Web interface	React, Vite, Tailwind CSS	Forms, portals, dashboards, and responsive navigation.
API	Python, FastAPI	Authentication, reporting, analysis, routing, and status updates.
Database and images	MongoDB Atlas and GridFS	Persistent report and account data plus uploaded images.
Map and charts	Leaflet / OpenStreetMap, Recharts	Issue locations, filters, and operational summaries.
Hosting	Vercel, Render, MongoDB Atlas	Static frontend, API service, and managed database.

Image analysis in the current app uses simple measurable features such as edge density, brightness, and color variation. Text categorization uses keyword matching. This is not a general-purpose trained vision model or an external generative AI service.

Duplicate reports can be clustered using distance and text similarity checks. The intended radius for nearby active reports is about 150 metres.

04 PRIVACY AND OPERATING LIMITS

Built for a small class demonstration

Privacy by design

- Guest reports can be submitted without creating an account or entering personal contact details.
- Citizen registration asks for a username and password. The demo should not use a real name, phone number, or personal email address.
- Public issue and tracking responses omit the reporter account object. The tracker searches by complaint ID, not by phone or email.
- Officer and administrator actions require an authenticated role. Demo persona switching also requires a private class access key.
- Report photos and location can still identify a person or home. Use fictional test data and avoid faces, number plates, and precise private addresses.

Deployment and limits

The current hosted demo uses Vercel for the frontend, Render for the API, and MongoDB Atlas for persistent data. Render free services may sleep when idle, so the first API request may take longer while the service wakes. Wait, then retry if a page reports a temporary load error.

The project is intended for a small classroom demonstration of about five simultaneous users. That is a planning target, not a verified capacity limit: the app has not been load-tested and has no production-scale service guarantee. Do not use it for real emergencies, official case management, or sensitive civic submissions.

The sample records are demonstration data. AI priority is a heuristic, not a real municipal dispatch decision. For a real deployment, the project would need formal security and privacy review, monitoring, backups, reliability testing, and clear operational ownership.

Current demonstration

The hosted site is https://civic-connect-ai-sigma.vercel.app/. The seeded test report CC-2026-9785 is labeled as test data and is currently In Progress. It can be used to show tracking without creating another report.